# Random Forest

⊛ A random Forest is a collection of trees. It predicts based on the majority vote.

Example: A Forest of 101 trees. 72 trees predict positive (survived), 29 predict negative (not Survived)

⇒ The forest predicts positive with certainty of $\frac{72}{101}$

⊛ We would like the trees in the forest are as different as possible, i.e. a diversed group of trees.

⊛ But how?

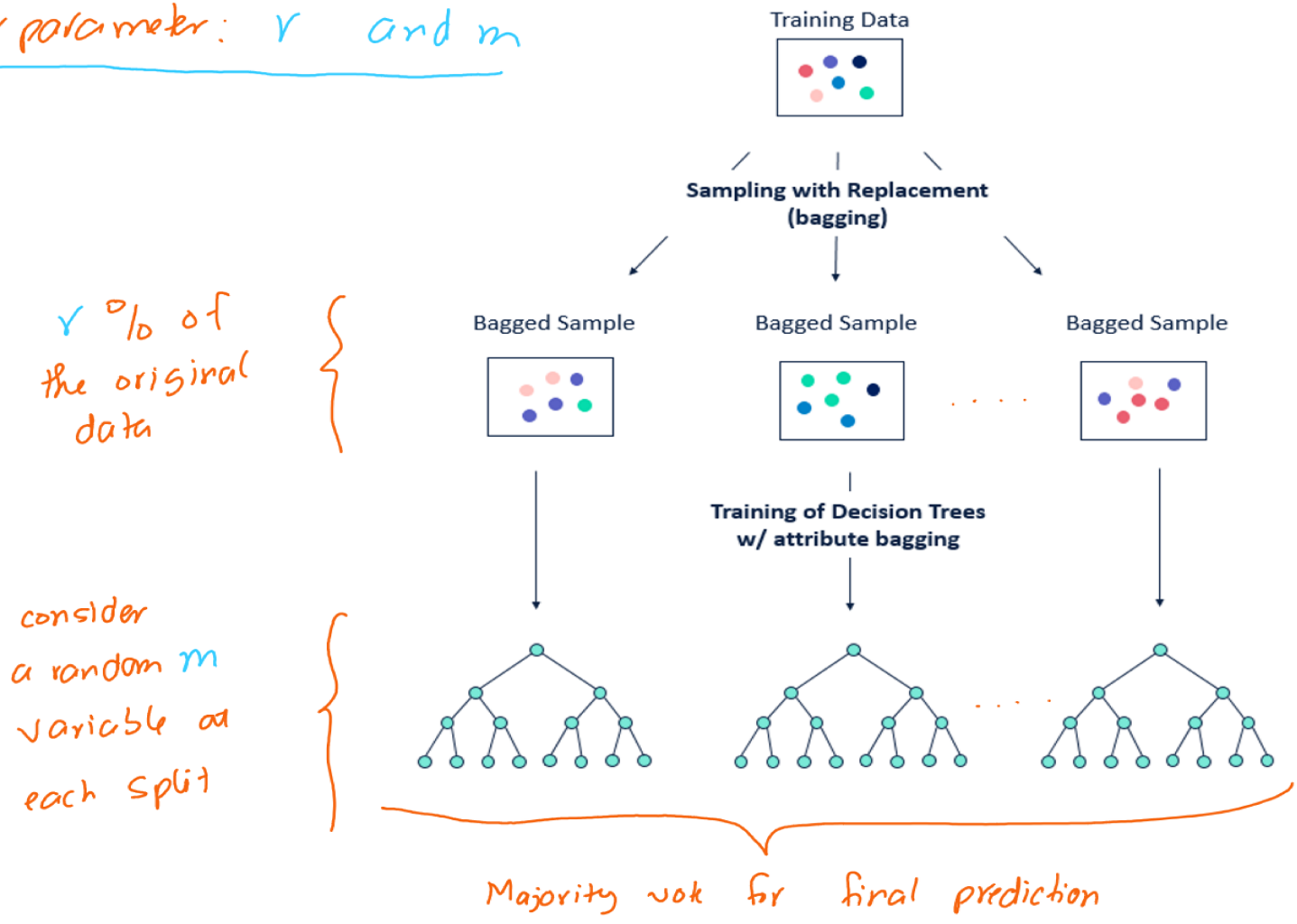
① we only use a random portion of the training data (40%) to train a tree

② At every split, only consider a random portion of variables (columns) to decide the best split.

Individually a tree in a forest is not as good as a regular tree but collectively the forest is "better" than a regular tree.

# Big Picture of Random Forest

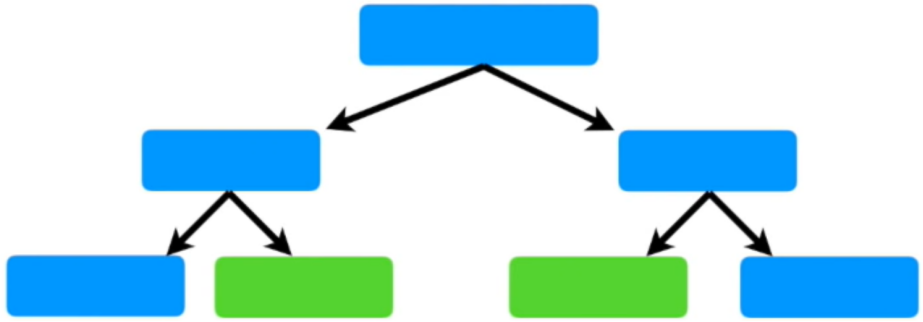
Hyper parameter: $r$ and $m$



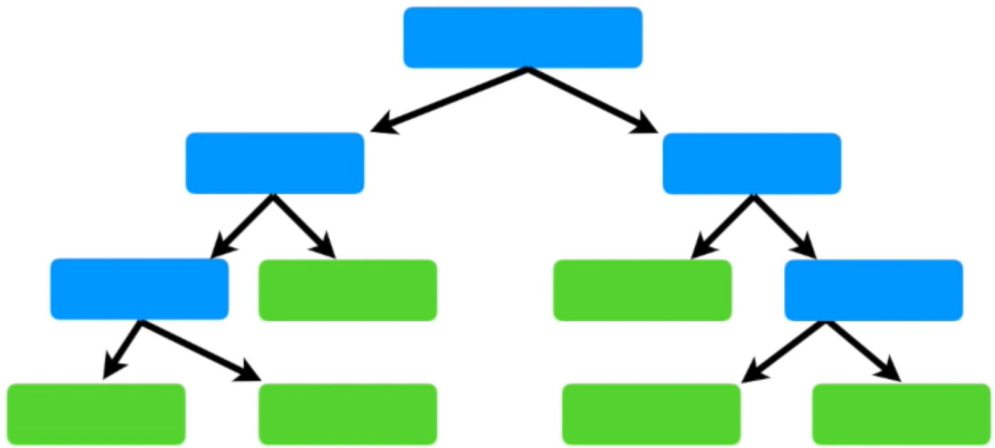
Random Forests are made out of decision trees

a stump is a tree with 1 split only

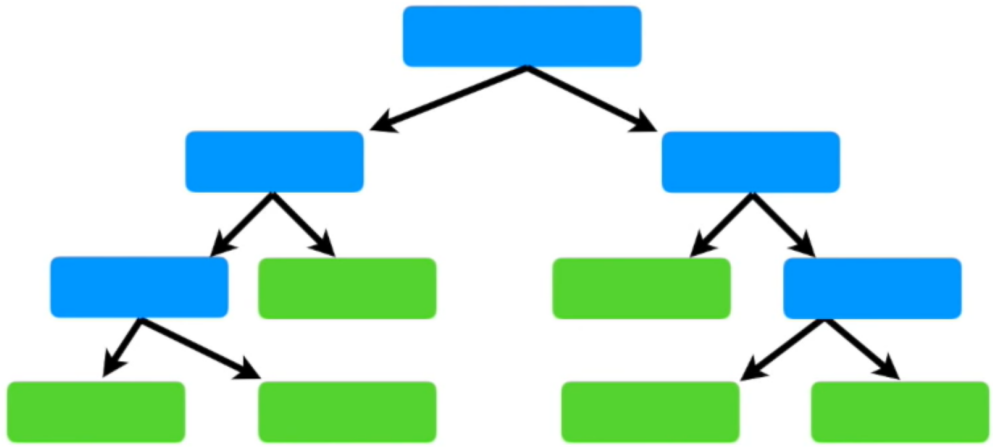
Random Forests are made out of decision trees



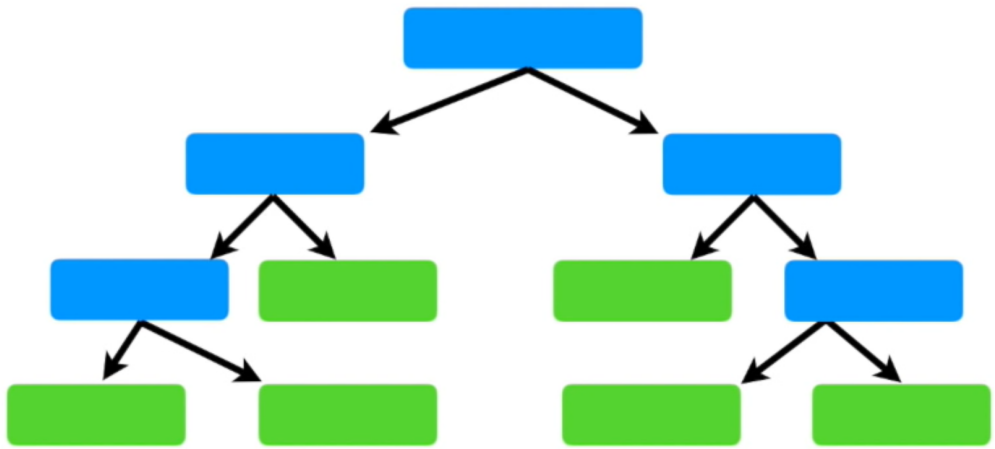
Random Forests are made out of decision trees



Decision Trees are easy to build, easy to use
and easy to interpret…

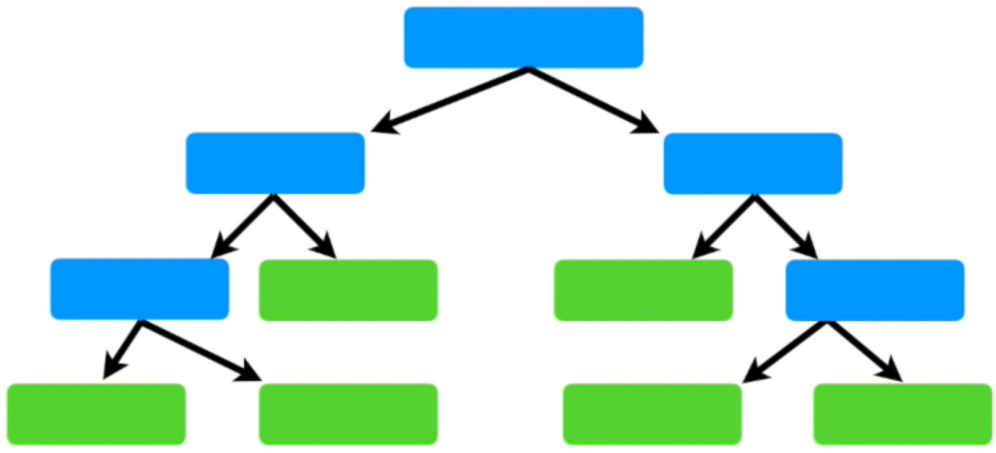


Decision Trees are easy to build, easy to use
and easy to interpret…

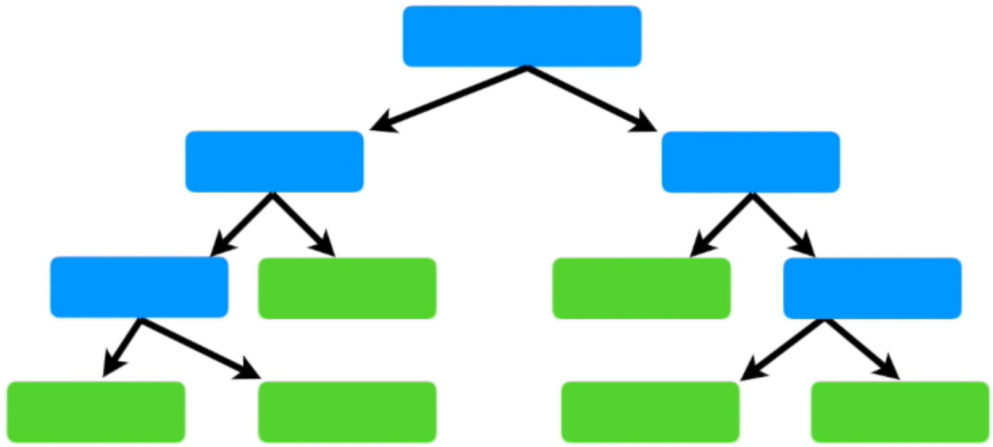


…but in practice they are not that awesome.

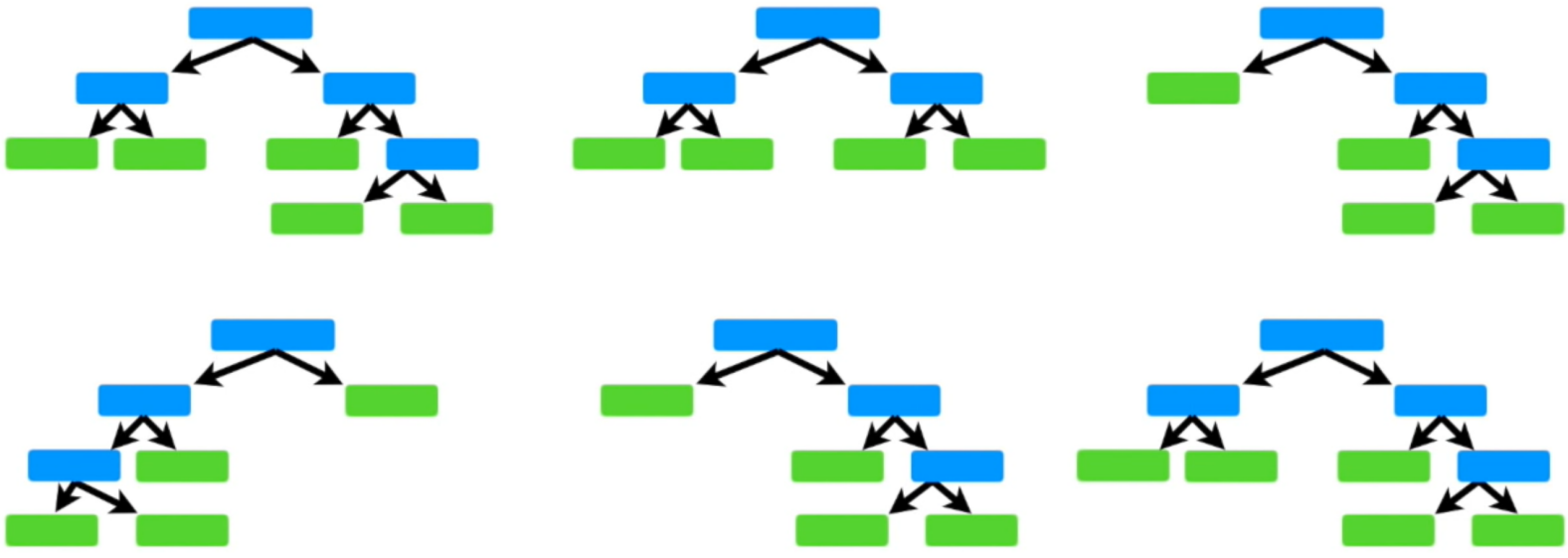
To quote from ***The Elements of Statistical Learning*** (aka The Bible of Machine Learning), “Trees have one aspect that prevents them from being the ideal tool for predictive learning, namely **inaccuracy**.”



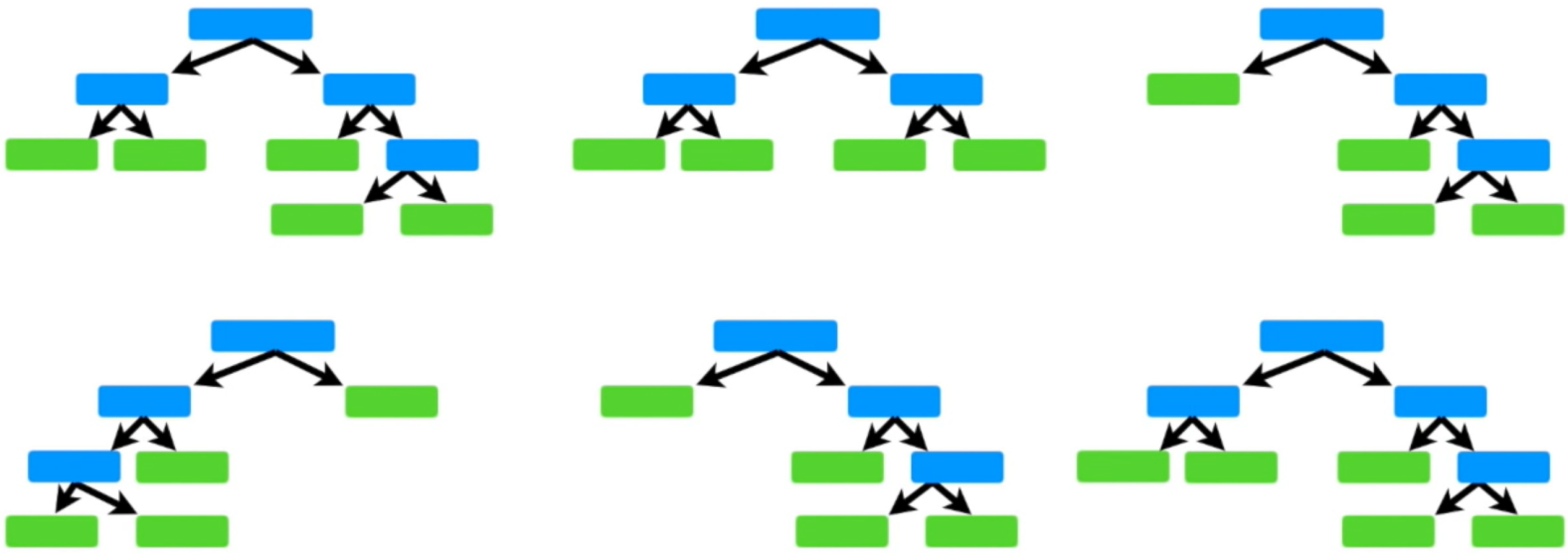
In other words, they work great with the data used to create them, but **they are not flexible when it comes to classifying new samples**.



The good news is that **Random Forests** combine the simplicity of decision trees with flexibility resulting in a vast improvement in accuracy.



So let’s make a Random Forest!!!



**Step 1:** Create a “bootstrapped” dataset.

Original Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| No | No | No | 125 | No |
| Yes | Yes | Yes | 180 | Yes |
| Yes | Yes | No | 210 | No |
| Yes | No | Yes | 167 | Yes |

Imagine that these 4 samples are the entire dataset that we are going to build a tree from…

Original Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| No | No | No | 125 | No |
| Yes | Yes | Yes | 180 | Yes |
| Yes | Yes | No | 210 | No |
| Yes | No | Yes | 167 | Yes |

Bootstrapped Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|

To create a bootstrapped dataset that is the same size as the original, we just randomly select samples from the original dataset.

Original Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| No | No | No | 125 | No |
| Yes | Yes | Yes | 180 | Yes |
| Yes | Yes | No | 210 | No |
| Yes | No | Yes | 167 | Yes |

Bootstrapped Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|

To create a bootstrapped dataset that is the same size as the original, we just randomly select samples from the original dataset.

The important detail is that we're allowed to pick the same sample more than once.

Original Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| No | No | No | 125 | No |
| Yes | Yes | Yes | 180 | Yes |
| Yes | Yes | No | 210 | No |
| Yes | No | Yes | 167 | Yes |

This is the first sample that we randomly select…

Bootstrapped Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|

Original Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| No | No | No | 125 | No |
| Yes | Yes | Yes | 180 | Yes |
| Yes | Yes | No | 210 | No |
| Yes | No | Yes | 167 | Yes |

…so it's the first sample in our bootstrapped dataset.

Bootstrapped Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | Yes | 180 | Yes |

Original Dataset

| Chest Pain | Good Blood Circ | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| No | No | No | 125 | No |
| Yes | Yes | Yes | 180 | Yes |
| Yes | Yes | No | 210 | No |
| Yes | No | Yes | 167 | Yes |

This is the second randomly selected sample from the original dataset…

Bootstrapped Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | Yes | 180 | Yes |

Original Dataset

| Chest Pain | Good Blood Circ | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| No | No | No | 125 | No |
| Yes | Yes | Yes | 180 | Yes |
| Yes | Yes | No | 210 | No |
| Yes | No | Yes | 167 | Yes |

…so it's the second sample in our bootstrapped dataset.

Bootstrapped Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | Yes | 180 | Yes |
| No | No | No | 125 | No |

Original Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| No | No | No | 125 | No |
| Yes | Yes | Yes | 180 | Yes |
| Yes | Yes | No | 210 | No |
| Yes | No | Yes | 167 | Yes |

Here's the third randomly selected sample…

Bootstrapped Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | Yes | 180 | Yes |
| No | No | No | 125 | No |

Original Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| No | No | No | 125 | No |
| Yes | Yes | Yes | 180 | Yes |
| Yes | Yes | No | 210 | No |
| Yes | No | Yes | 167 | Yes |

…so here it is in the bootstrapped dataset.

Bootstrapped Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | Yes | 180 | Yes |
| No | No | No | 125 | No |
| Yes | No | Yes | 167 | Yes |

Original Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| No | No | No | 125 | No |
| Yes | Yes | Yes | 180 | Yes |
| Yes | Yes | No | 210 | No |
| Yes | No | Yes | 167 | Yes |

Bootstrapped Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | Yes | 180 | Yes |
| No | No | No | 125 | No |
| Yes | No | Yes | 167 | Yes |

Here's the fourth randomly selected sample (**Note**: it's the same as the third)…

Original Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| No | No | No | 125 | No |
| Yes | Yes | Yes | 180 | Yes |
| Yes | Yes | No | 210 | No |
| Yes | No | Yes | 167 | Yes |

Bootstrapped Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | Yes | 180 | Yes |
| No | No | No | 125 | No |
| Yes | No | Yes | 167 | Yes |
| Yes | No | Yes | 167 | Yes |

...and here it is.

We've created a bootstrapped dataset!!!

## Original Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| No | No | No | 125 | No |
| Yes | Yes | Yes | 180 | Yes |
| Yes | Yes | No | 210 | No |
| Yes | No | Yes | 167 | Yes |

## Bootstrapped Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | Yes | 180 | Yes |
| No | No | No | 125 | No |
| Yes | No | Yes | 167 | Yes |
| Yes | No | Yes | 167 | Yes |

**Step 2:** Create a decision tree using the bootstrapped dataset, but only use a random subset of variables (or columns) at each step.

Bootstrapped Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | Yes | 180 | Yes |
| No | No | No | 125 | No |
| Yes | No | Yes | 167 | Yes |
| Yes | No | Yes | 167 | Yes |

**Step 2:** Create a decision tree using the bootstrapped dataset, but only use a random subset of variables (or columns) at each step.

In this example, we will only consider 2 variables (columns) at each step.

Bootstrapped Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | Yes | 180 | Yes |
| No | No | No | 125 | No |
| Yes | No | Yes | 167 | Yes |
| Yes | No | Yes | 167 | Yes |

**Step 2:** Create a decision tree using the bootstrapped dataset, but only use a random subset of variables (or columns) at each step.

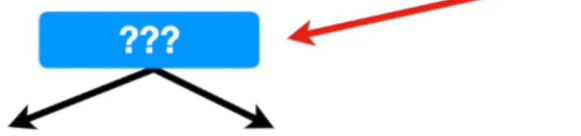
In this example, we will only consider 2 variables (columns) at each step.

**NOTE:** We'll talk more about how to determine the optimal number of variables to consider later…

Bootstrapped Dataset

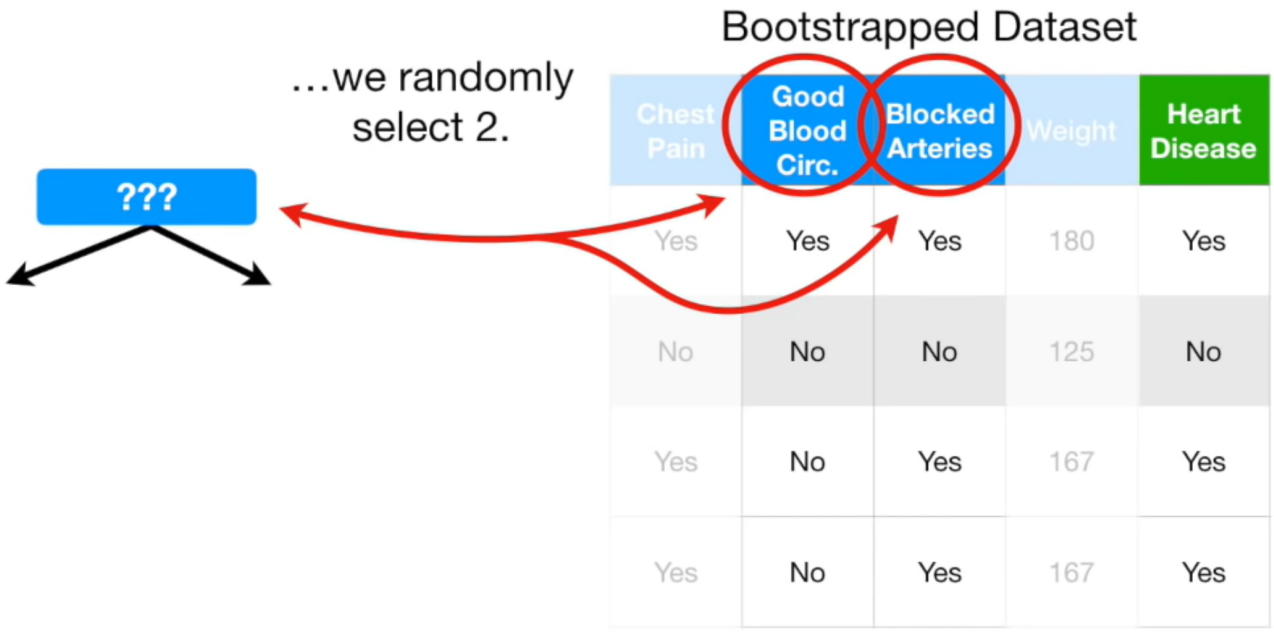
| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | Yes | 180 | Yes |
| No | No | No | 125 | No |
| Yes | No | Yes | 167 | Yes |
| Yes | No | Yes | 167 | Yes |

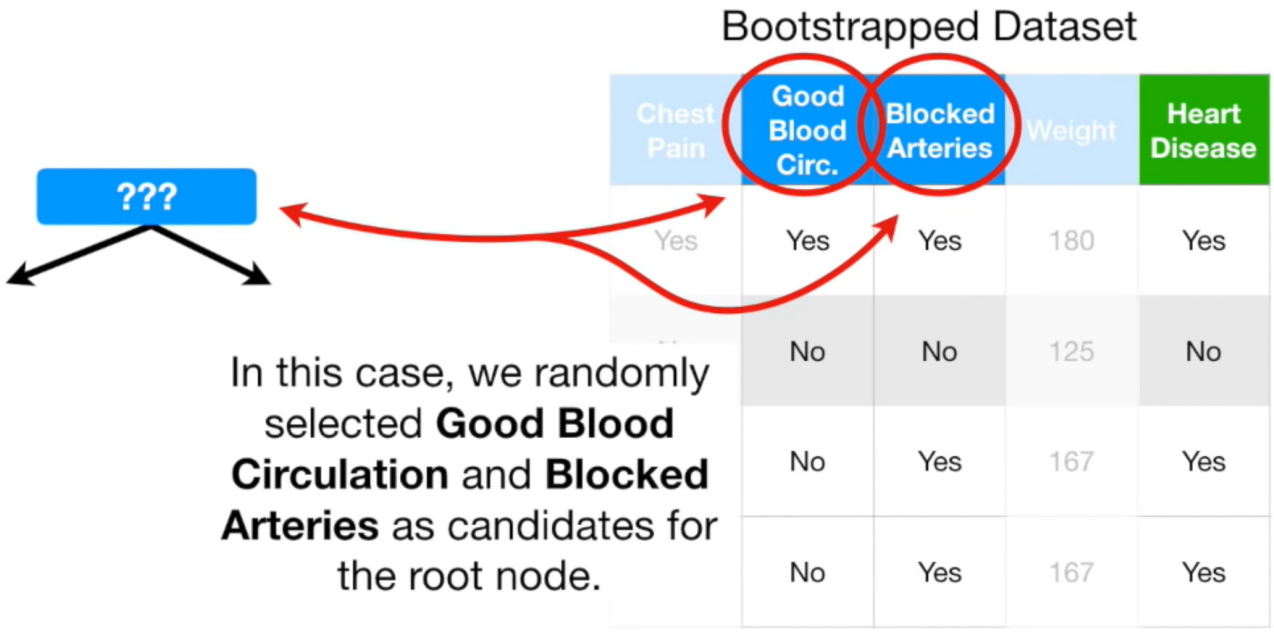
Thus, instead of considering all 4 variables to figure out how to split the root node…




Bootstrapped Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | Yes | 180 | Yes |
| No | No | No | 125 | No |
| Yes | No | Yes | 167 | Yes |
| Yes | No | Yes | 167 | Yes |

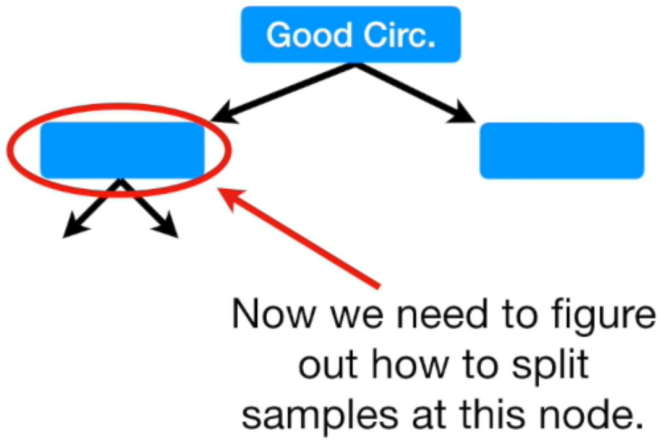

Bootstrapped Dataset
…we randomly select 2.
???
Chest Pain
Good Blood Circ.
Blocked Arteries
Weight
Heart Disease
Yes
Yes
Yes
180
Yes
No
No
No
125
No
Yes
No
Yes
167
Yes
Yes
No
Yes
167
Yes


Bootstrapped Dataset
???
Chest Pain
Good Blood Circ.
Blocked Arteries
Weight
Heart Disease
Yes
Yes
Yes
180
Yes
No
No
125
No
No
Yes
167
Yes
No
Yes
167
Yes
In this case, we randomly selected **Good Blood Circulation** and **Blocked Arteries** as candidates for the root node.

Just for the sake of the example, assume that **Good Blood Circulation** did the best job separating the samples.

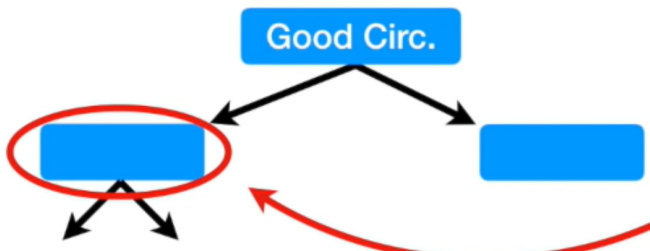
Bootstrapped Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | Yes | 180 | Yes |
| No | No | No | 125 | No |
| Yes | No | Yes | 167 | Yes |
| Yes | No | Yes | 167 | Yes |




Bootstrapped Dataset

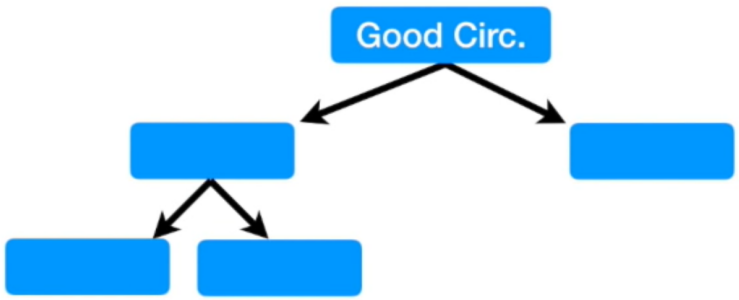
| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | Yes | 180 | Yes |
| No | No | No | 125 | No |
| Yes | No | Yes | 167 | Yes |
| Yes | No | Yes | 167 | Yes |




Just like for the root, we randomly select 2 variables as candidates, instead of all 4 columns.

Bootstrapped Dataset

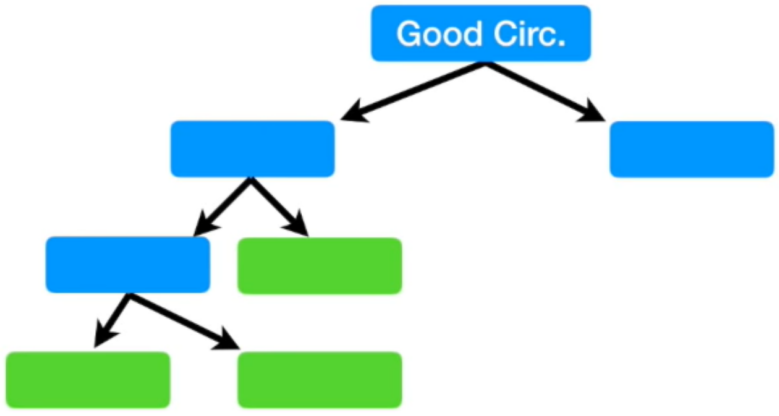
| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | Yes | 180 | Yes |
| No | No | No | 125 | No |
| Yes | No | Yes | 167 | Yes |
| Yes | No | Yes | 167 | Yes |




Bootstrapped Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | Yes | 180 | Yes |
| Yes | No | Yes | 167 | Yes |
| Yes | No | Yes | 167 | Yes |
| Yes | No | Yes | 167 | Yes |
| Yes | No | Yes | 167 | Yes |

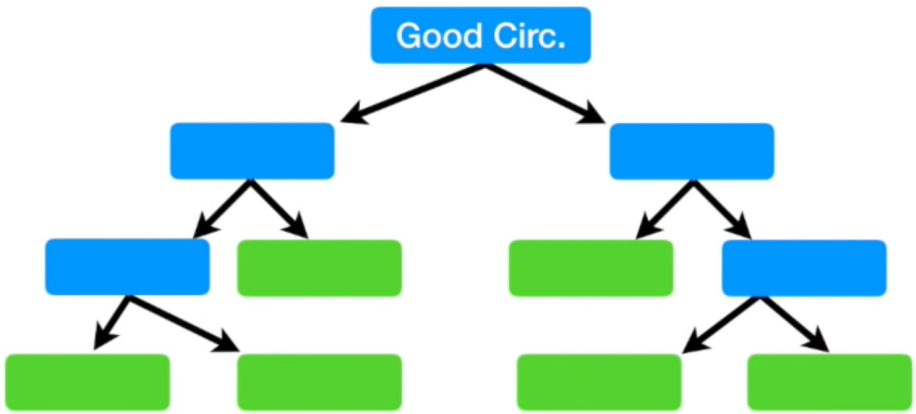
And we just build the tree as usual, but only considering a random subset of variables at each step.




Bootstrapped Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | Yes | 180 | Yes |
| Yes | No | Yes | 167 | Yes |
| Yes | No | Yes | 167 | Yes |
| Yes | No | Yes | 167 | Yes |
| Yes | No | Yes | 167 | Yes |

And we just build the tree as usual, but only considering a random subset of variables at each step.

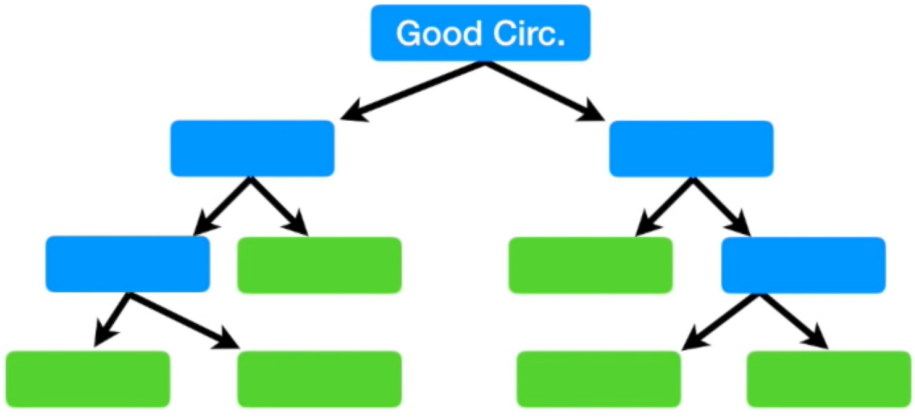



Bootstrapped Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | Yes | 180 | Yes |
| No | No | No | 125 | No |
| Yes | No | Yes | 167 | Yes |
| Yes | No | Yes | 167 | Yes |

We built a tree…

1) Using a bootstrapped dataset



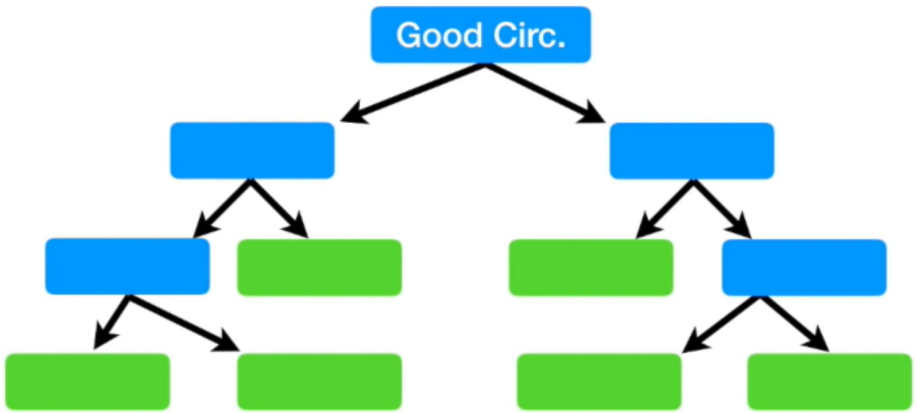

Bootstrapped Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | Yes | 180 | Yes |
| No | No | No | 125 | No |
| Yes | No | Yes | 167 | Yes |
| Yes | No | Yes | 167 | Yes |

We built a tree…

1) Using a bootstrapped dataset

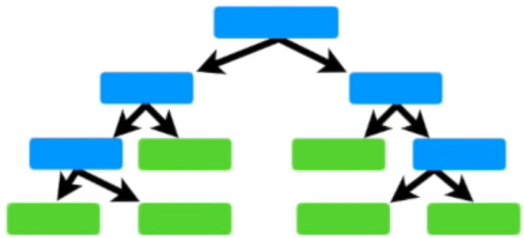
2) Only considering a random a subset of variables at each step.



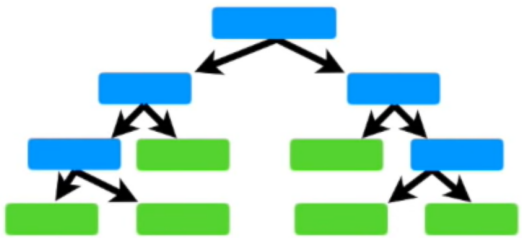

Bootstrapped Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | Yes | 180 | Yes |
| No | No | No | 125 | No |
| Yes | No | Yes | 167 | Yes |
| Yes | No | Yes | 167 | Yes |

Here’s the tree we just made…



**Now go back to Step 1 and repeat:** Make a new bootstrapped dataset and build a tree considering a subset of variables at each step.



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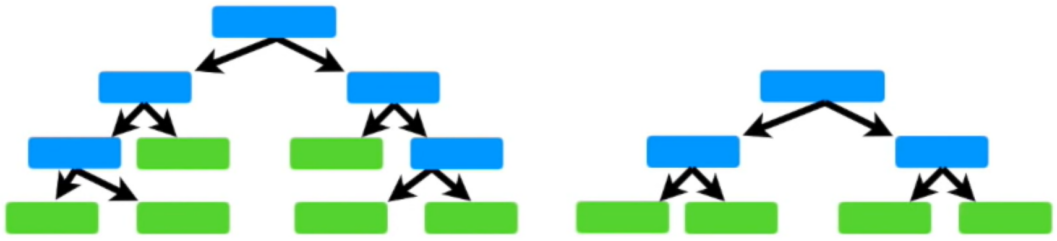

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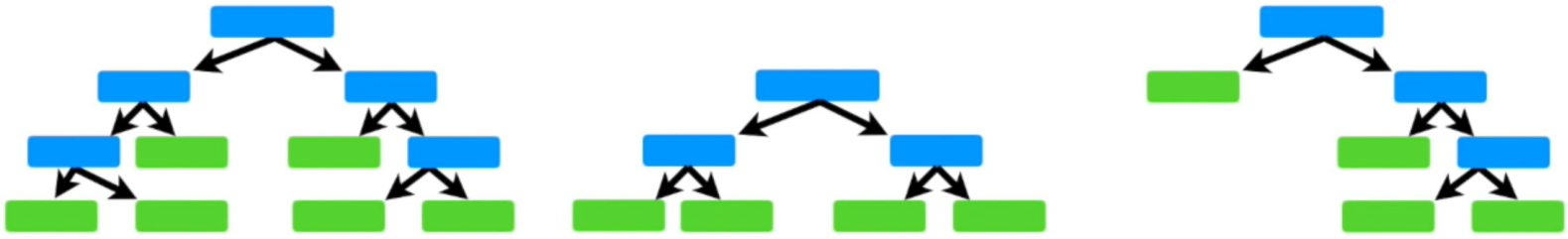

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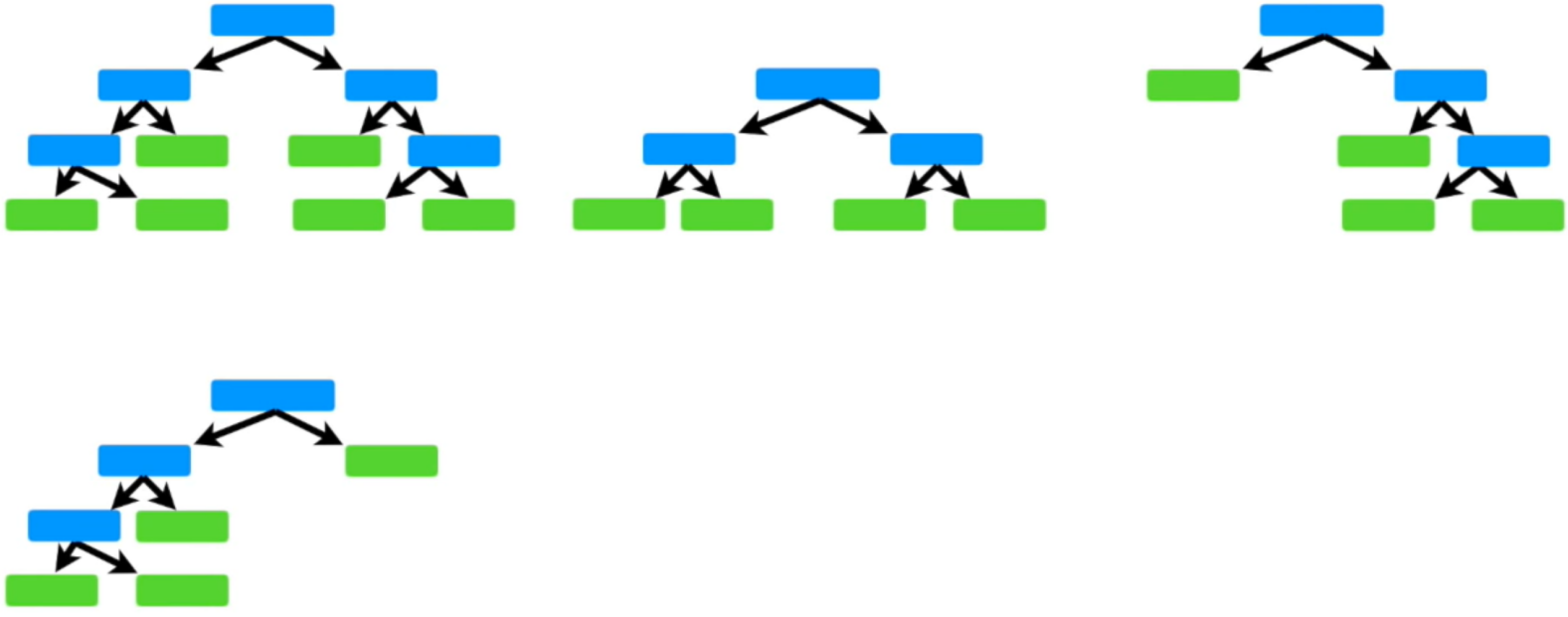
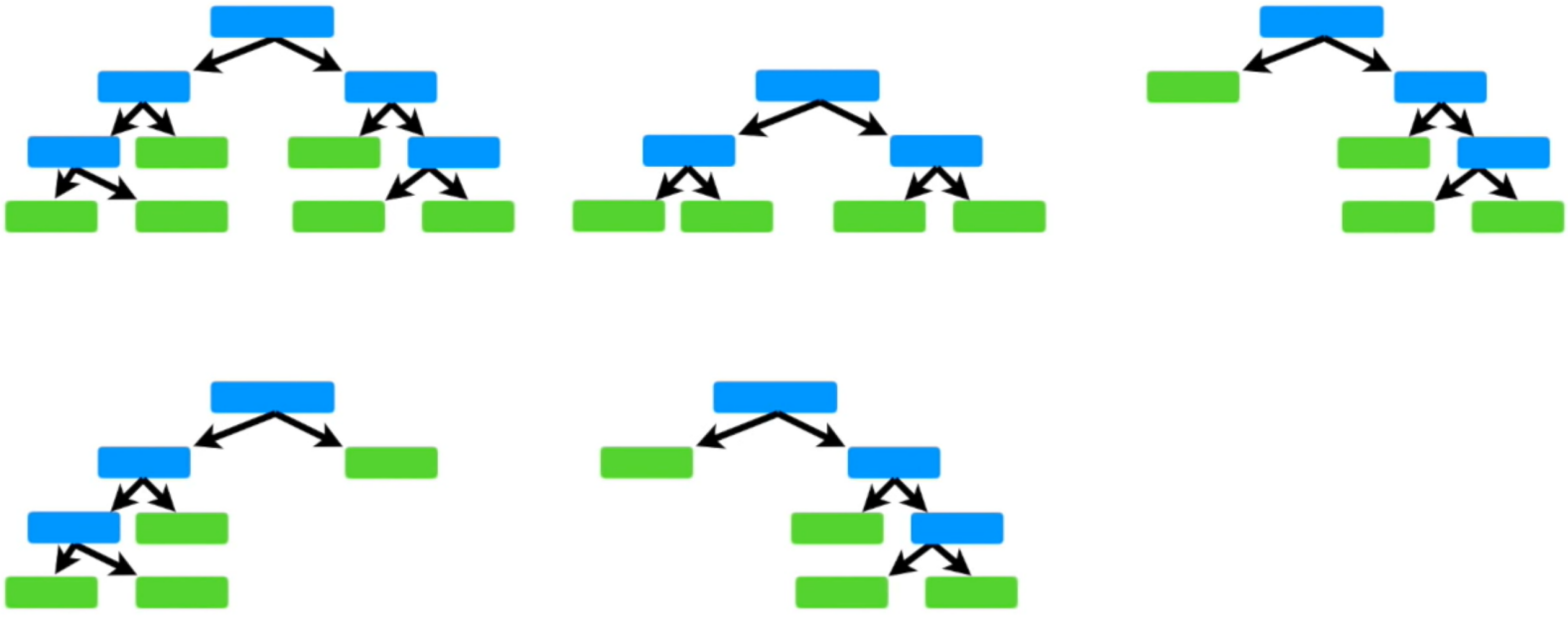
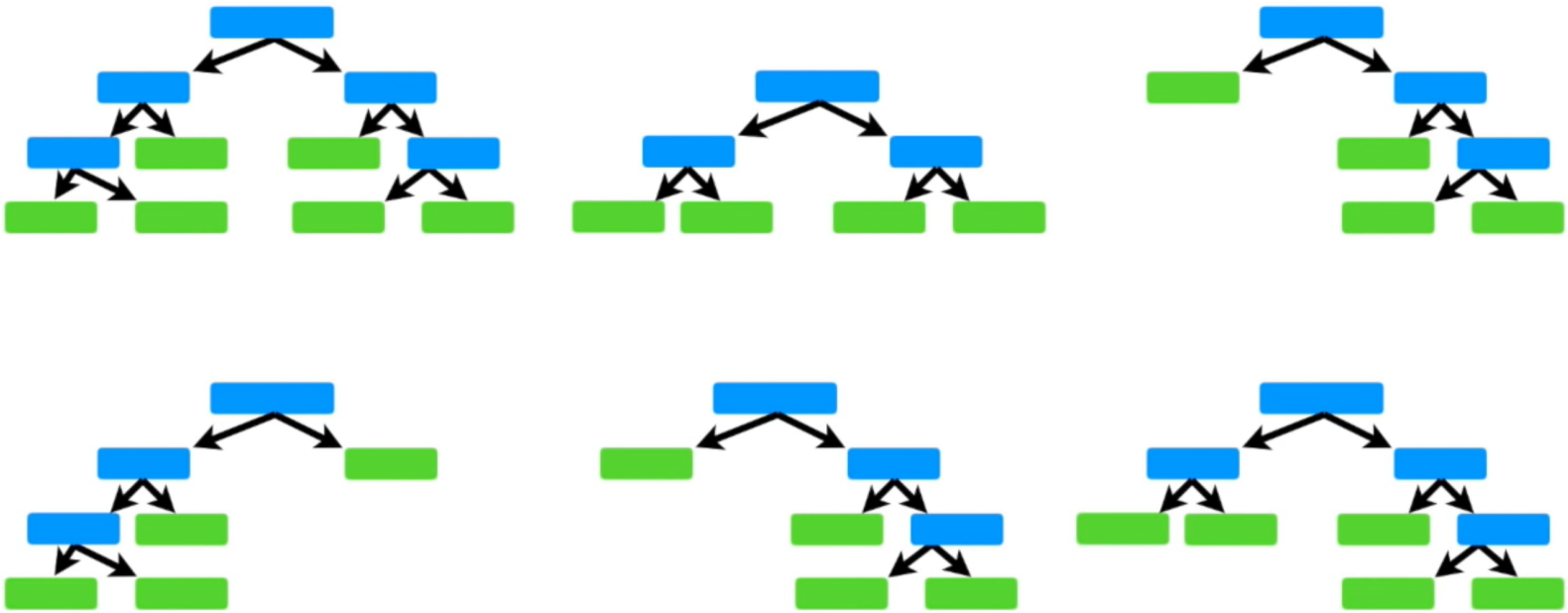


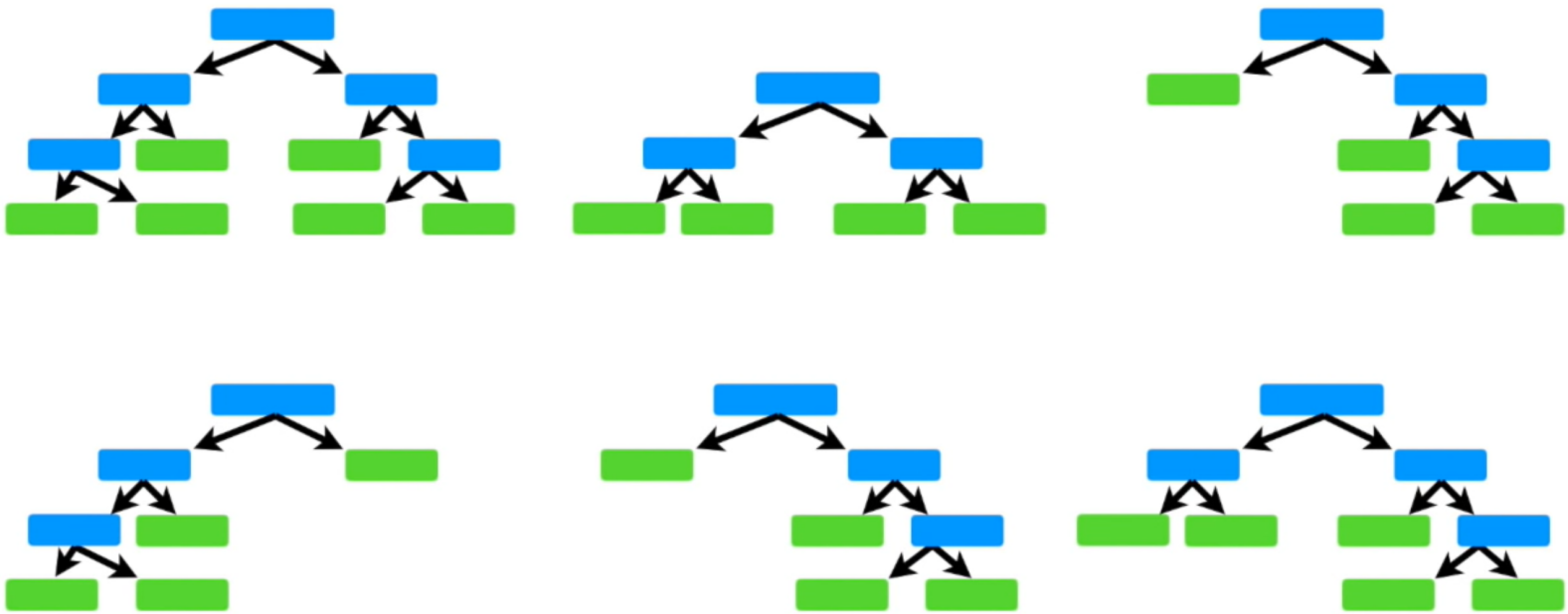
**Now go back to Step 1 and repeat:** Make a new bootstrapped dataset and build a tree considering a subset of variables at each step.



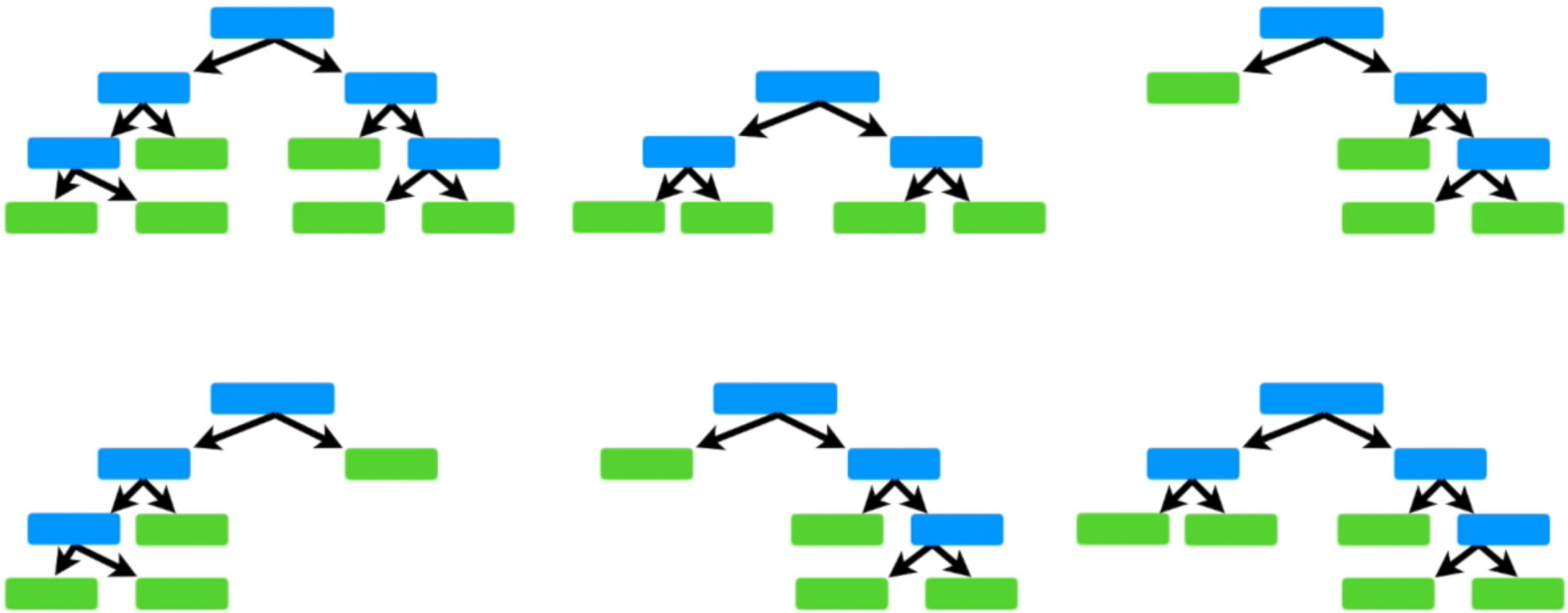
**Now go back to Step 1 and repeat:** Make a new bootstrapped dataset and build a tree considering a subset of variables at each step.



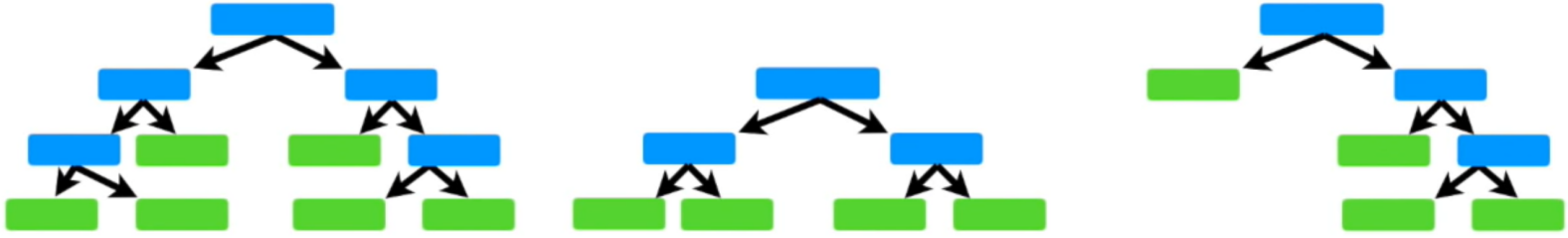
Ideally, you'd do this 100's of times, but we only have space to show 6… but you get the idea.



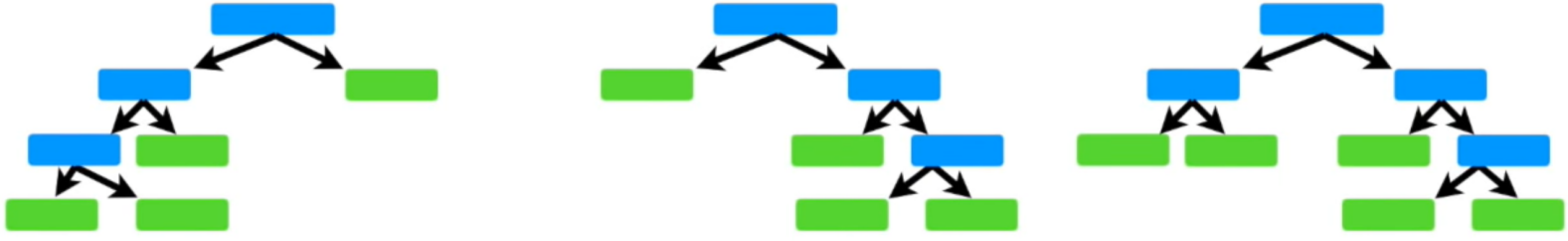
Using a bootstrapped sample and considering only a subset of the variables at each step results in a wide variety of trees.



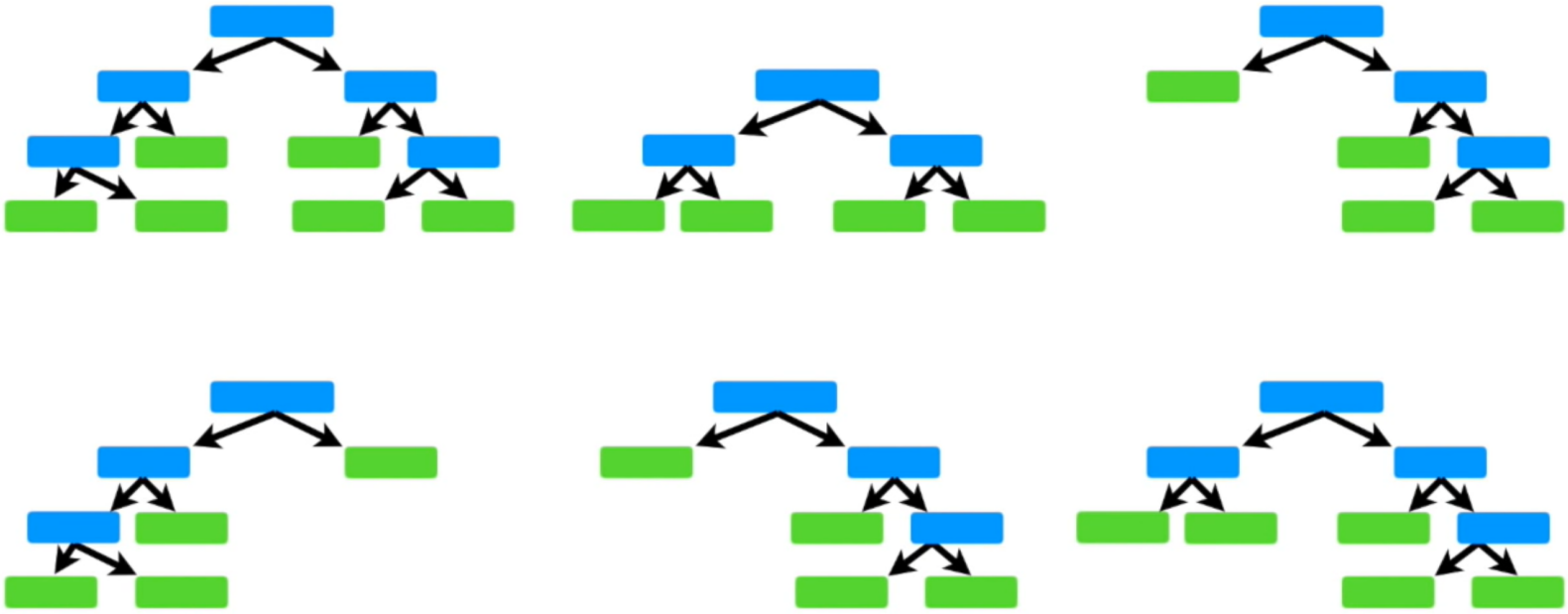
Using a bootstrapped sample and considering only a subset of the variables at each step results in a wide variety of trees.



The variety is what makes random forests more effective than individual decision trees.



Sweet!!! Now that we've created a random forest, how do we use it?



Well, first we get a new patient…

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | No | No | 168 | |

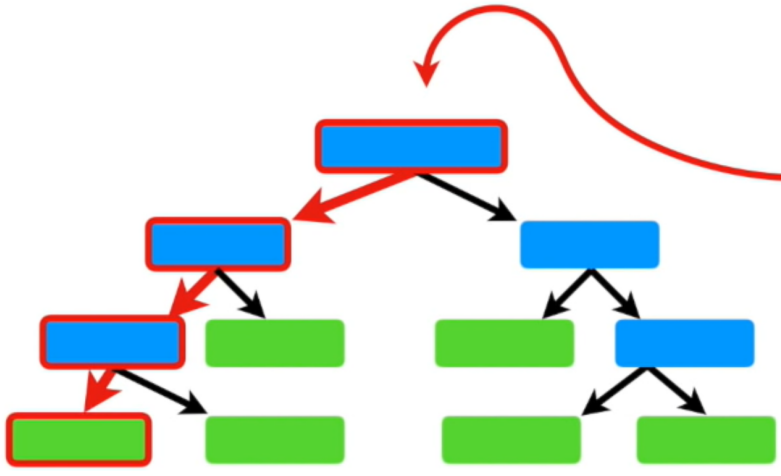
| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | No | No | 168 | |

…we've got all the measurements…

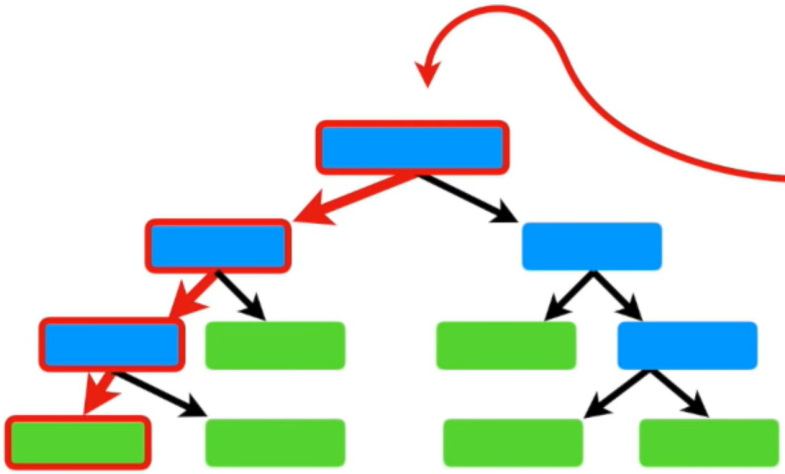
| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | No | No | 168 | |

…and now we want to know if they have heart disease or not.

So we take the data and run it down the first tree that we made…

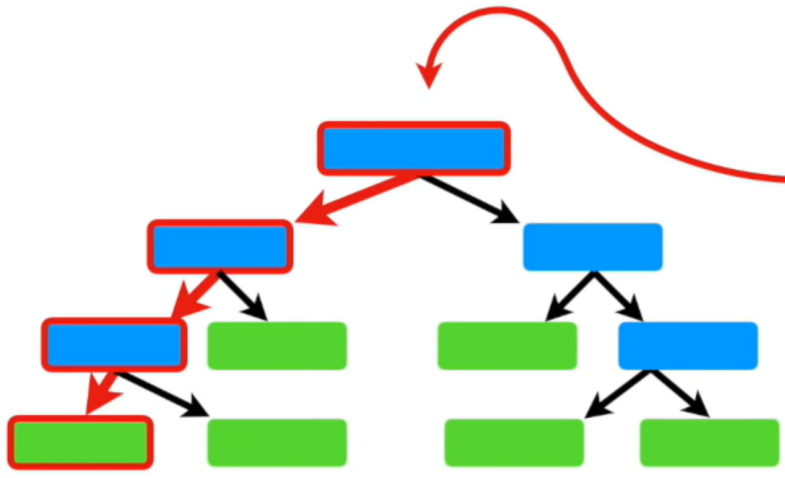


| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | No | No | 168 | |



| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | No | No | 168 | |

The first tree says
"Yes"…



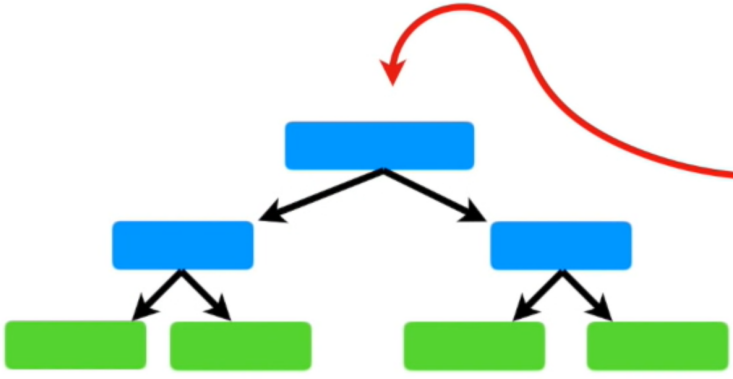
| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | No | No | 168 | |

| Heart Disease | |
|---|---|
| Yes | No |
| 1 | 0 |

The first tree says "Yes"…

…and we keep track of that here.

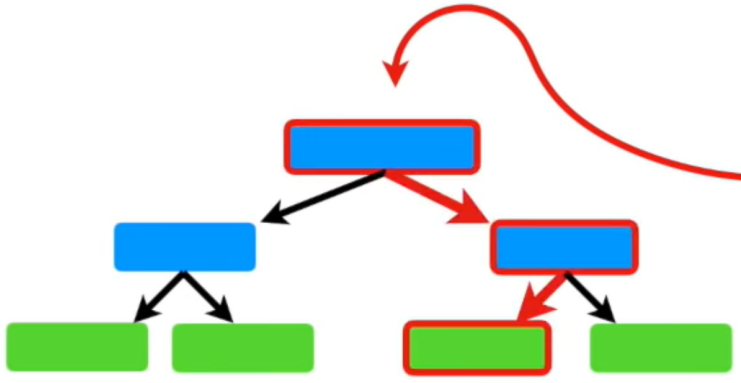
Now we run the data down the second tree that we made…



| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | No | No | 168 | |

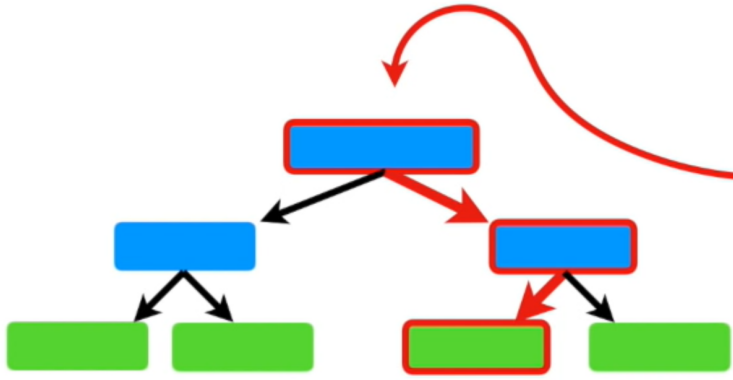
| Heart Disease | |
|---|---|
| Yes | No |
| 1 | 0 |

Now we run the data down the second tree that we made…



| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | No | No | 168 | |

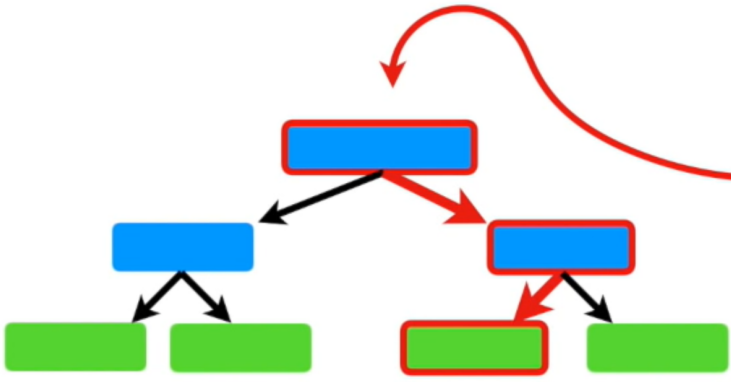
| Heart Disease | |
|---|---|
| Yes | No |
| 1 | 0 |



The second tree also says “Yes”…

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | No | No | 168 | |

| Heart Disease | |
|---|---|
| Yes | No |
| 1 | 0 |



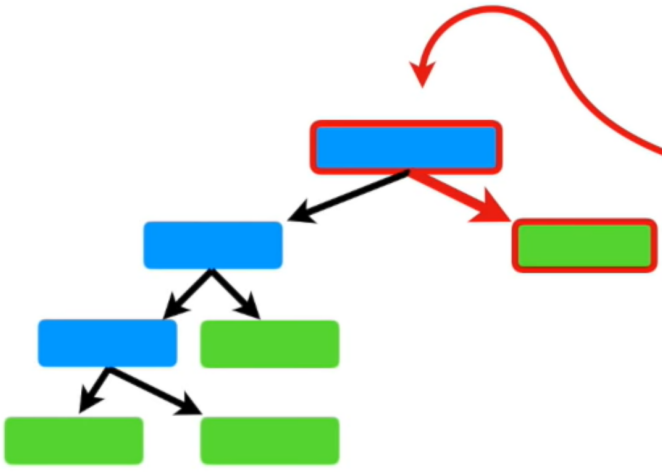
| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | No | No | 168 | |

The second tree also says “Yes”…

| Heart Disease | |
|---|---|
| Yes | No |
| 2 | 0 |

…and we keep track of that here.

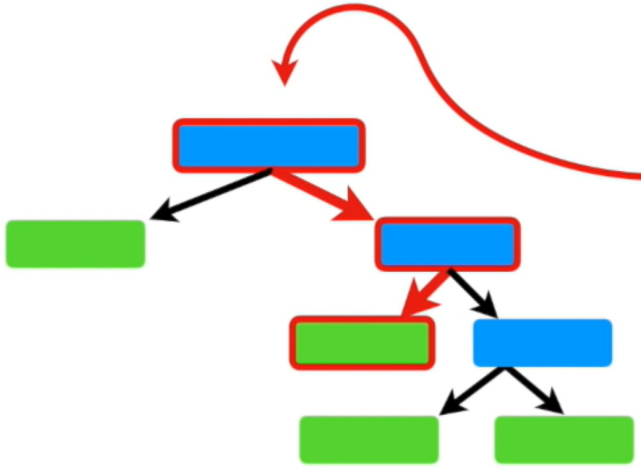
Then we repeat for all the trees that we made…



| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | No | No | 168 | |

| Heart Disease | |
|---|---|
| Yes | No |
| 3 | 1 |

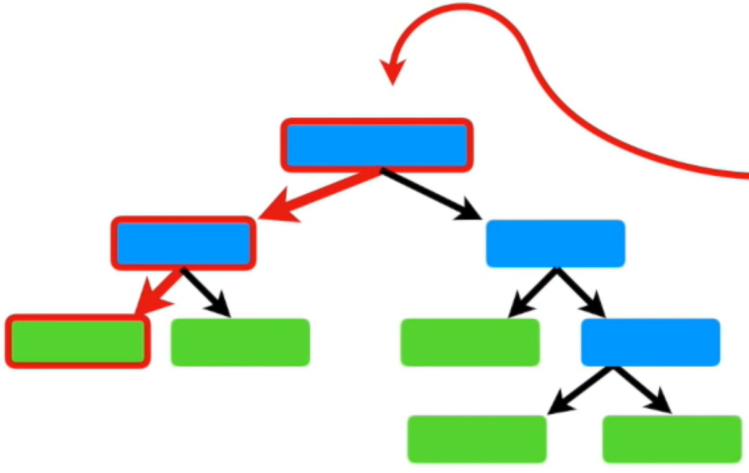
Then we repeat for all the trees that we made…



| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | No | No | 168 | |

| Heart Disease | |
|---|---|
| Yes | No |
| 4 | 1 |

Then we repeat for all the trees that we made…



| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | No | No | 168 | |

| Heart Disease | |
|---|---|
| Yes | No |
| 5 | 1 |

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | No | No | 168 | |

After running the data down all of the trees in the random forest, we see which option received more votes.

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | No | No | 168 | **YES** |

In this case, "**Yes**" received the most votes, so we will conclude that this patient has heart disease.

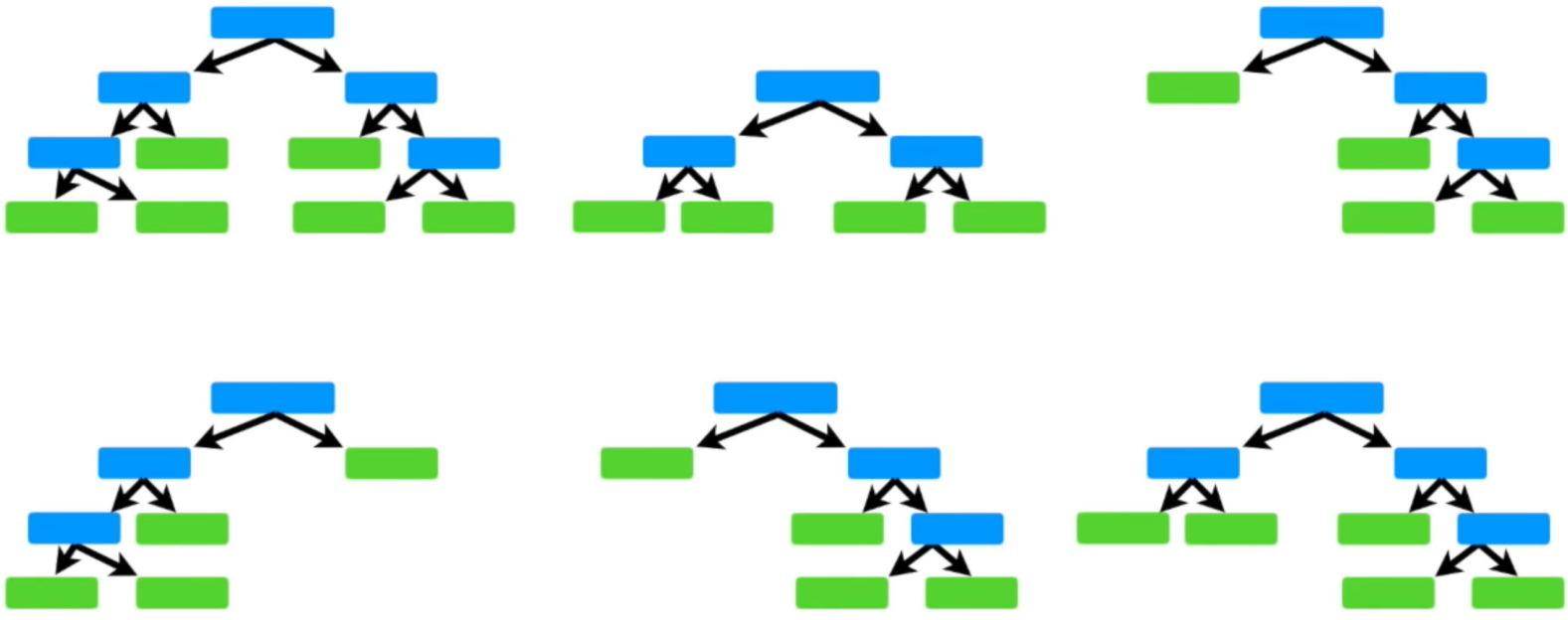
# Terminology Alert!!!

**B**ootstrapping the data plus using the **agg**regate to make a decision is called "**Bagging**"

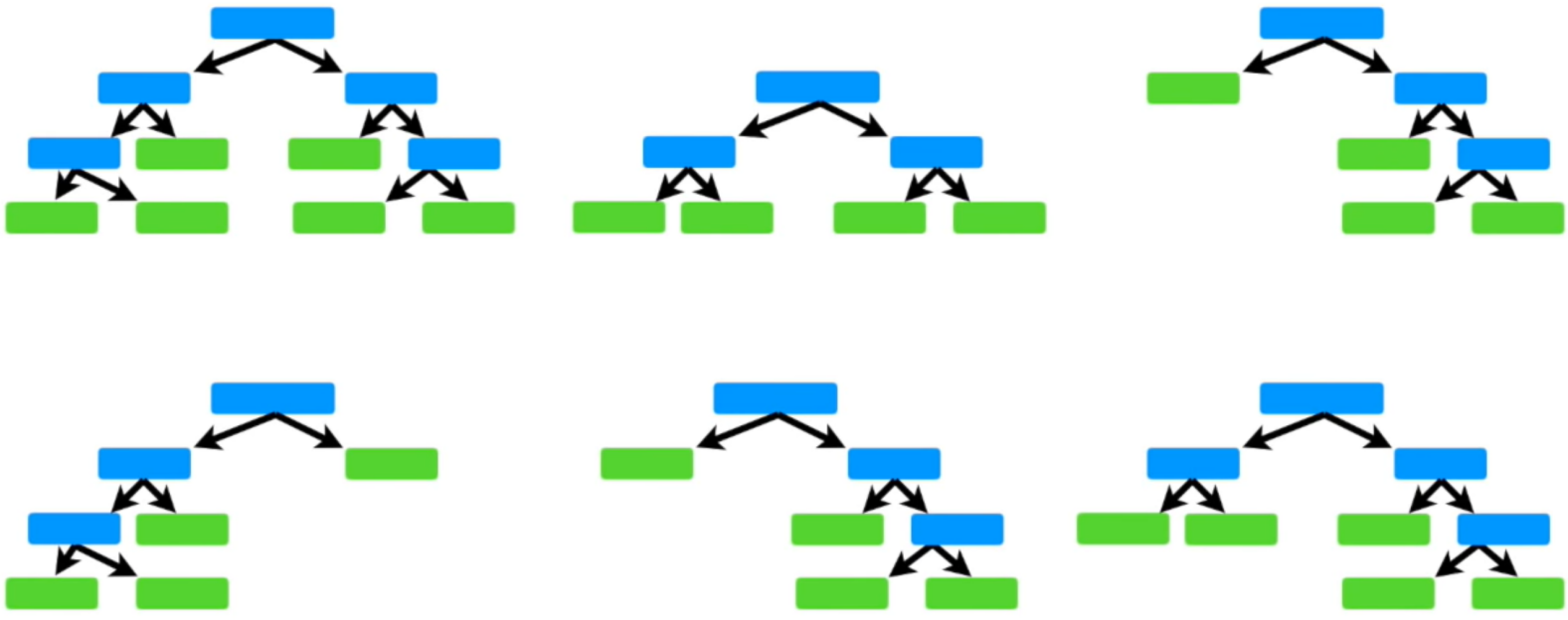
| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | No | No | 168 | **YES** |

| Heart Disease | |
|---|---|
| Yes | No |
| 5 | 1 |

OK, now we've seen how to create and use a random forest…



How do we know if it's any good?



| | Tree 1 | Tree 2 | Tree 3 | Tree 4 | Tree 5 | Tree 6 | Tree 7 | True Target | out-of-bag predict | |
|---|---|---|---|---|---|---|---|---|---|---|
| ① | V | N | N | V | V | P | V | P | N | wrong |
| ② | V | V | V | P | P | V | V | P | P | correct |
| ③ | V | P | V | V | V | V | P | N | P | wrong |
| ④ | N | V | N | N | V | N | N | N | N | correct |
| ⑤ | P | V | P | V | N | V | V | N | P | wrong |
| ⑥ | V | N | V | V | N | P | N | N | N | correct |

oob misclassification rate $= \frac{3}{6} = 50\%$

Training data for tree 1

| |
|---|
| ① |
| ① |
| ② |
| ③ |
| ③ |
| ⑥ |

bag 1

④ ⑤

out-of-bag data

⊛ Hyperparameters of Forest.
(Tuning parameter)

① The number of trees.

② The number of variables considered at each split in trees: The lower the number, the more diversed the forest. By default, it is $\sqrt{d}$, where $d$ is the total number of variables

# Out-of-bag errors

- Random Forest do not need a validation dataset!

Remember when we created the bootstrapped dataset?

Original Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| No | No | No | 125 | No |
| Yes | Yes | Yes | 180 | Yes |
| Yes | Yes | No | 210 | No |
| Yes | No | Yes | 167 | Yes |

Bootstrapped Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|

We allowed duplicate entries in the bootstrapped dataset…

Original Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| No | No | No | 125 | No |
| Yes | Yes | Yes | 180 | Yes |
| Yes | Yes | No | 210 | No |
| Yes | No | Yes | 167 | Yes |

Bootstrapped Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | Yes | 180 | Yes |
| No | No | No | 125 | No |
| Yes | No | Yes | 167 | Yes |
| Yes | No | Yes | 167 | Yes |

As a result, this entry was not included in the bootstrapped dataset.

Original Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| No | No | No | 125 | No |
| Yes | Yes | Yes | 180 | Yes |
| Yes | Yes | No | 210 | No |
| Yes | No | Yes | 167 | Yes |

Bootstrapped Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | Yes | 180 | Yes |
| No | No | No | 125 | No |
| Yes | No | Yes | 167 | Yes |
| Yes | No | Yes | 167 | Yes |

Typically, about 1/3 of the original data does not end up in the bootstrapped dataset.

Original Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| No | No | No | 125 | No |
| Yes | Yes | Yes | 180 | Yes |
| Yes | Yes | No | 210 | No |
| Yes | No | Yes | 167 | Yes |

Bootstrapped Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | Yes | 180 | Yes |
| No | No | No | 125 | No |
| Yes | No | Yes | 167 | Yes |
| Yes | No | Yes | 167 | Yes |

Original Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| No | No | No | 125 | No |
| Yes | Yes | Yes | 180 | Yes |
| Yes | Yes | No | 210 | No |
| Yes | No | Yes | 167 | Yes |

Here is the entry that didn't end up in the bootstrapped dataset..

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | No | 210 | No |

Original Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| No | No | No | 125 | No |
| Yes | Yes | Yes | 180 | Yes |
| Yes | Yes | No | 210 | No |
| Yes | No | Yes | 167 | Yes |

(psst… if the original dataset were larger, we'd have more than just 1 entry over here…)

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | No | 210 | No |

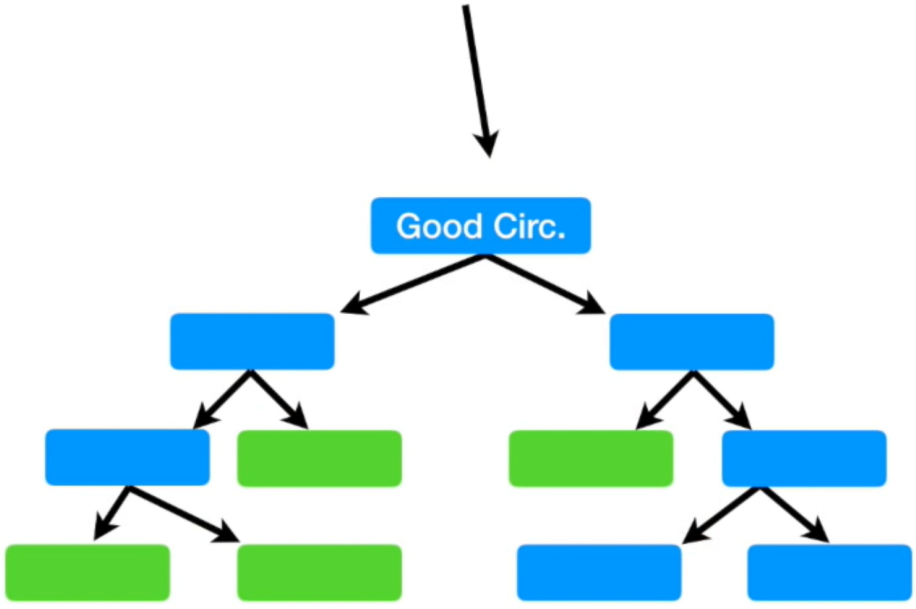
Original Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| No | No | No | 125 | No |
| Yes | Yes | Yes | 180 | Yes |
| Yes | Yes | No | 210 | No |
| Yes | No | Yes | 167 | Yes |

This is called the
"**Out-Of-Bag Dataset**"

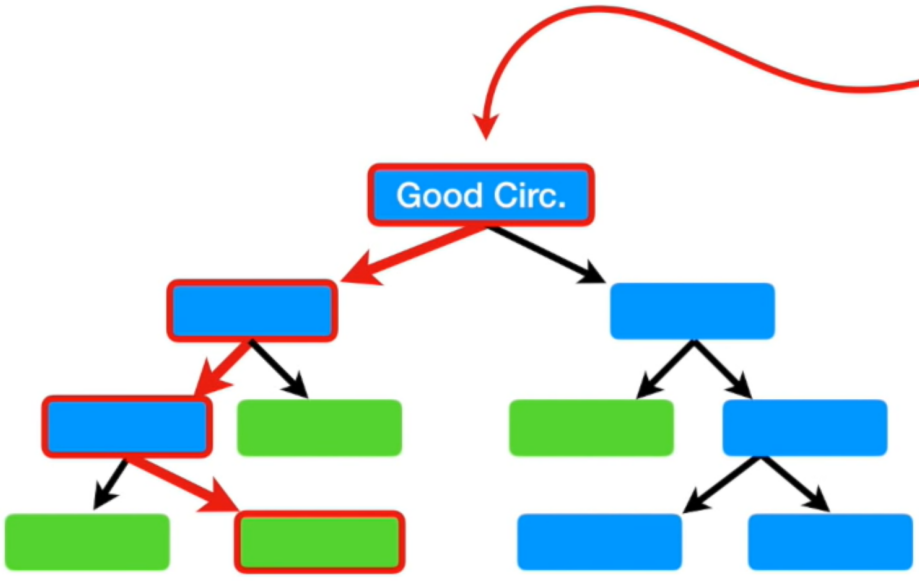
| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | No | 210 | No |

Since the Out-Of-Bag data was not used to create this tree…

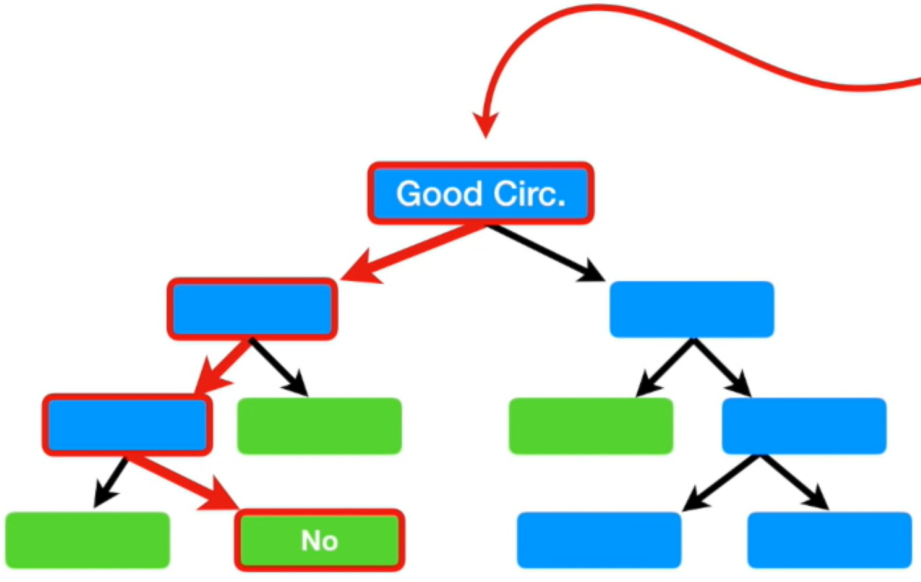



| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | No | 210 | No |

…we can run it through and see if it correctly classifies the sample as "No Heart Disease"

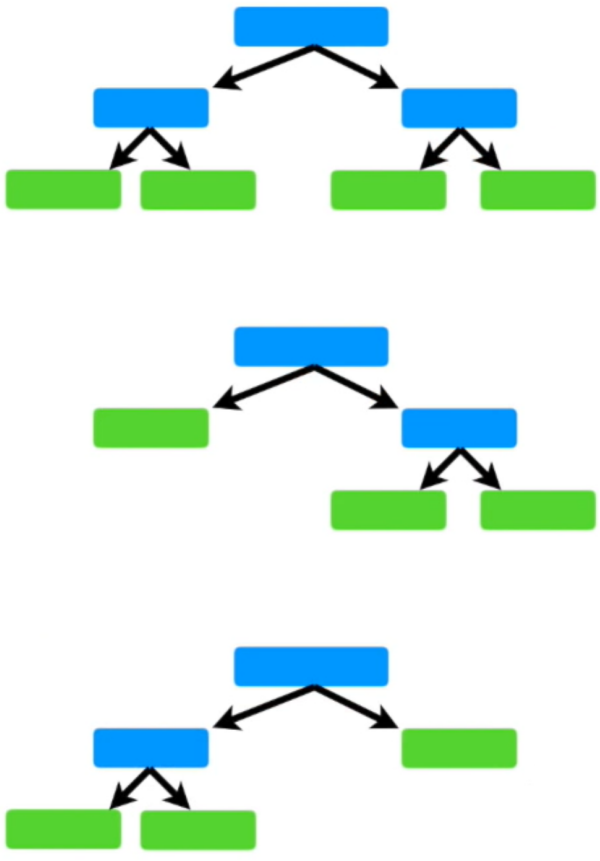



| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | No | 210 | No |



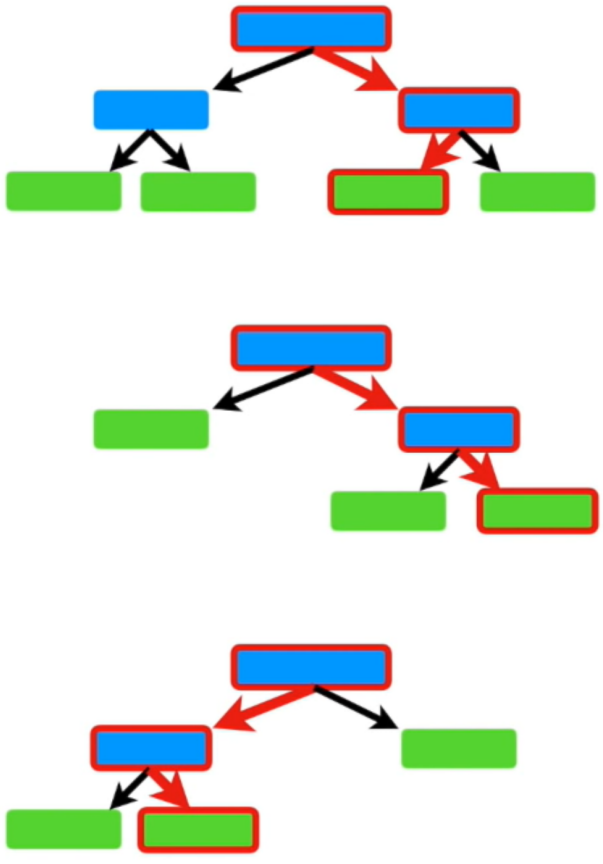

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | No | 210 | No |

In this case, the tree correctly labels the Out-of-Bag sample "**No**".



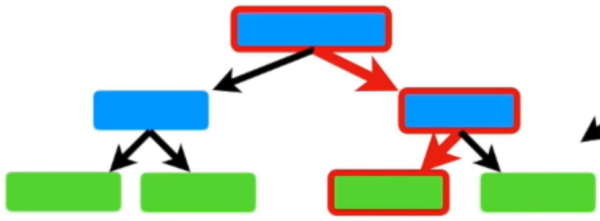
Then we run this Out-Of-Bag sample through all of the other trees that were built without it…

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | No | 210 | No |

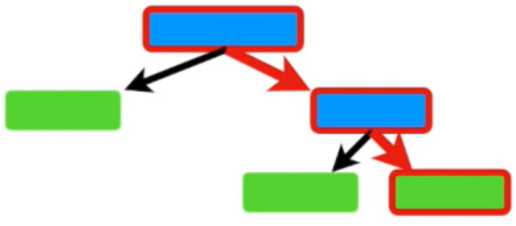


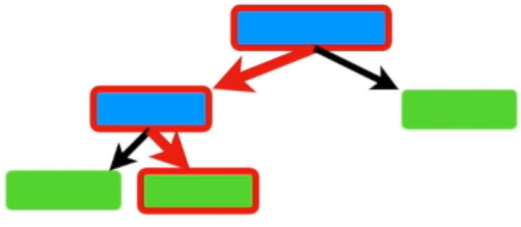
Then we run this Out-Of-Bag sample through all of the other trees that were built without it…

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | No | 210 | No |

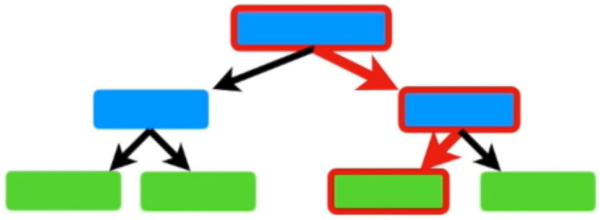


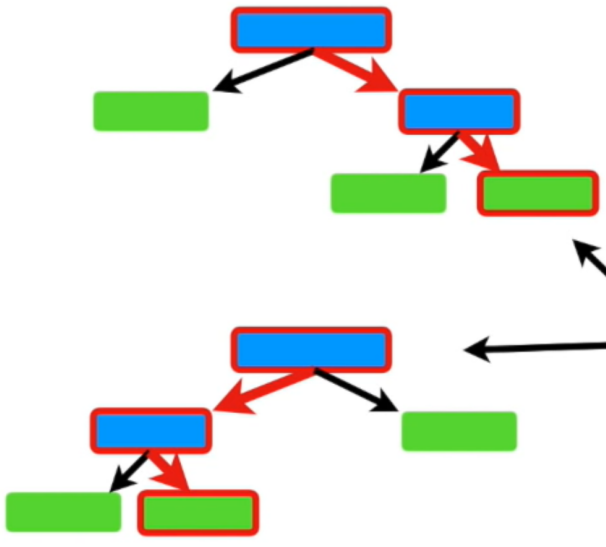
This tree incorrectly labeled the Out-of-Bag sample "**Yes**".





| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | No | 210 | No |



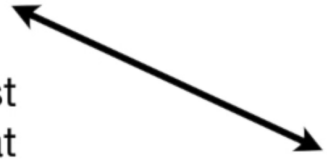


These trees correctly labeled the Out-of-Bag sample “**No**”.

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | No | 210 | No |

| Classification of the Out-Of-Bag sample | |
|---|---|
| Yes | No |
| 1 | 3 |

Since the label with the most votes wins, it is the label that we assign this Out-of-Bag sample.



| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | No | 210 | No |

| Classification of the Out-Of-Bag sample | |
|---|---|
| Yes | No |
| 1 | 3 |

Since the label with the most votes wins, it is the label that we assign this Out-of-Bag sample.

In this case, the Out-of-Bag sample is correctly labeled by the Random Forest.

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | No | 210 | No |

| Classification of the Out-Of-Bag sample | |
|---|---|
| Yes | No |
| 1 | 3 |

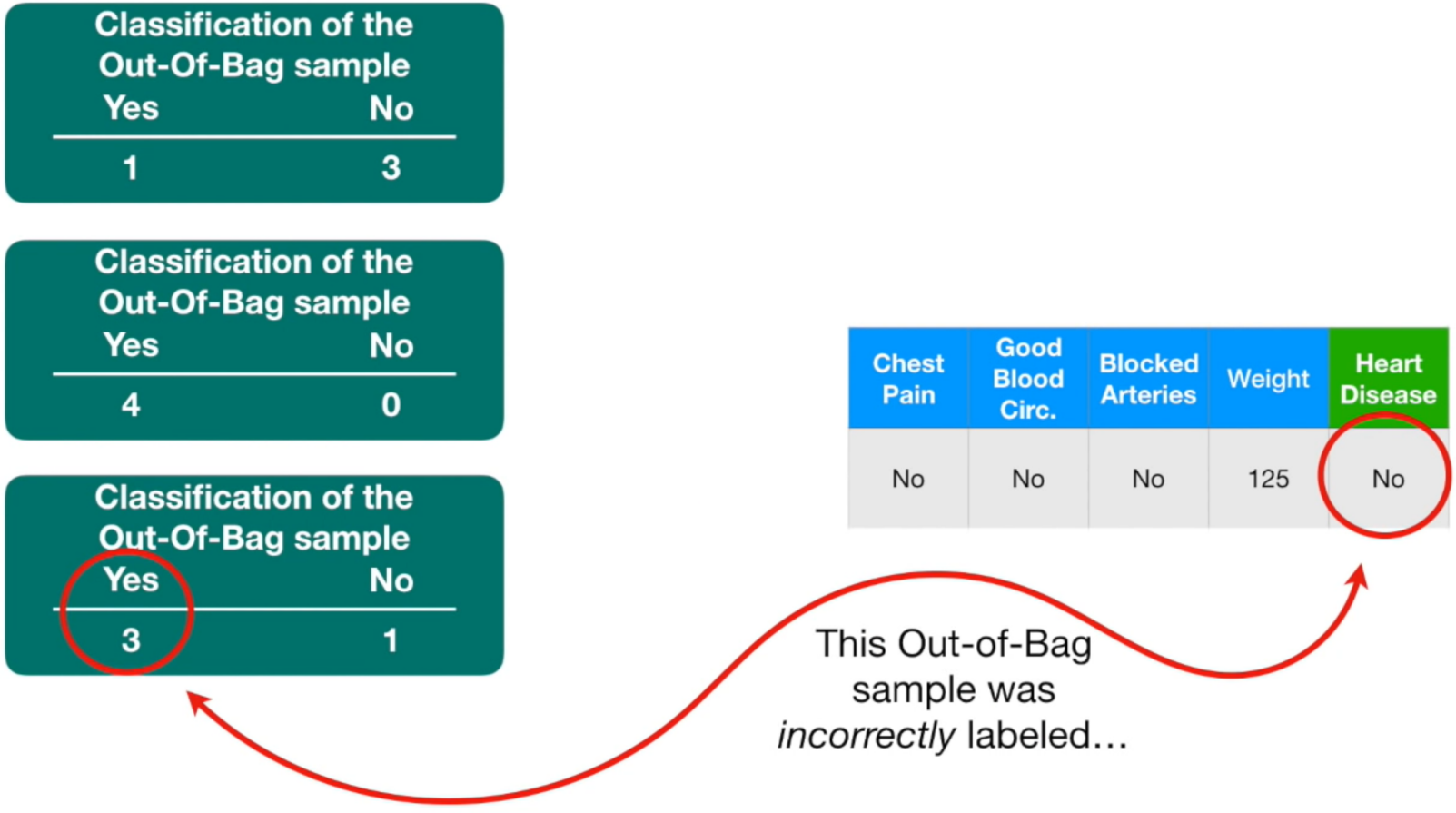
We then do the same thing for all of the other Out-Of-Bag samples for all of the trees.

| Classification of the Out-Of-Bag sample | |
|---|---|
| Yes | No |
| 1 | 3 |

| Classification of the Out-Of-Bag sample | |
|---|---|
| Yes | No |
| 4 | 0 |

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | Yes | 180 | Yes |

This Out-of-Bag sample was also correctly labeled…


Classification of the Out-Of-Bag sample
Yes
No
1
3
Classification of the Out-Of-Bag sample
Yes
No
4
0
Classification of the Out-Of-Bag sample
Yes
No
3
1
Chest Pain
Good Blood Circ.
Blocked Arteries
Weight
Heart Disease
No
No
No
125
No
This Out-of-Bag sample was *incorrectly* labeled…

| Classification of the Out-Of-Bag sample | |
|---|---|
| Yes | No |
| 1 | 3 |

| Classification of the Out-Of-Bag sample | |
|---|---|
| Yes | No |
| 4 | 0 |

| Classification of the Out-Of-Bag sample | |
|---|---|
| Yes | No |
| 3 | 1 |

etc… etc… etc…

Ultimately, we can measure how accurate our random forest is by the proportion of Out-Of-Bag samples that were correctly classified by the Random Forest.

| Classification of the Out-Of-Bag sample | |
|---|---|
| Yes | No |
| 1 | 3 |

| Classification of the Out-Of-Bag sample | |
|---|---|
| Yes | No |
| 4 | 0 |

| Classification of the Out-Of-Bag sample | |
|---|---|
| Yes | No |
| 3 | 1 |

etc… etc… etc…

Ultimately, we can measure how accurate our random forest is by the proportion of Out-Of-Bag samples that were correctly classified by the Random Forest.

The proportion of Out-Of-Bag samples that were *incorrectly* classified is the "**Out-Of-Bag Error**"

OK, we now know how to:

1) Build a Random Forest

2) Use a Random Forest

3) Estimate the accuracy of a Random Forest.

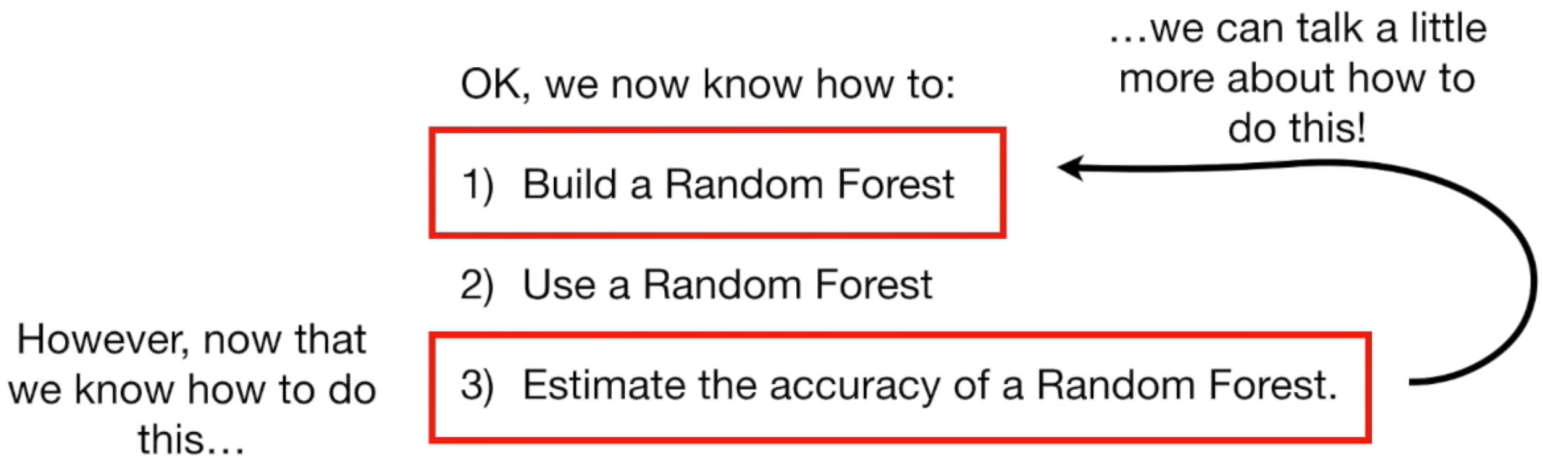
However, now that we know how to do this…

OK, we now know how to:

1) Build a Random Forest

2) Use a Random Forest

3) Estimate the accuracy of a Random Forest.


…we can talk a little more about how to do this!
OK, we now know how to:
1) Build a Random Forest
2) Use a Random Forest
3) Estimate the accuracy of a Random Forest.
However, now that we know how to do this…

Remember when we built our first tree and we only used 2 variables (columns of data) to make a decision at each step?

Bootstrapped Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | Yes | 180 | Yes |
| No | No | No | 125 | No |
| Yes | No | Yes | 167 | Yes |
| Yes | No | Yes | 167 | Yes |

Now we can compare the Out-Of-Bag error for a random forest built using only 2 variables per step…

Bootstrapped Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | Yes | 180 | Yes |
| No | No | No | 125 | No |
| Yes | No | Yes | 167 | Yes |
| Yes | No | Yes | 167 | Yes |

…to random forest built using 3 variables per step…

Bootstrapped Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | Yes | 180 | Yes |
| No | No | No | 125 | No |
| Yes | No | Yes | 167 | Yes |
| Yes | No | Yes | 167 | Yes |

…to random forest built using 3 variables per step…

…and we test a bunch of different settings and choose the most accurate random forest.

Bootstrapped Dataset

| Chest Pain | Good Blood Circ. | Blocked Arteries | Weight | Heart Disease |
|---|---|---|---|---|
| Yes | Yes | Yes | 180 | Yes |
| No | No | No | 125 | No |
| Yes | No | Yes | 167 | Yes |
| Yes | No | Yes | 167 | Yes |

In other words…

1) Build a Random Forest

In other words…

1) Build a Random Forest

2) Estimate the accuracy of a Random Forest.

In other words…

…change the number of variables used per step…

1) Build a Random Forest

2) Estimate the accuracy of a Random Forest.

In other words…

…change the number of variables used per step…

1) Build a Random Forest

2) Estimate the accuracy of a Random Forest.

Do this for a bunch of times and then choose the one that is most accurate.

In other words…

…change the number of variables used per step…

1) Build a Random Forest

2) Estimate the accuracy of a Random Forest.

Typically, we start by using the square of the number of variables and then try a a few settings above and below that value.

Titanic data

| | Age | Sex | . . |
|---|---|---|---|
| : | - | . | ' , |
| : | | . | - . |
| . | - | | |
| . | . | , | . |

Images Data